

A Written Report

CYBER SECURITY

Submitted to

Course : CSE 1102

Section : 10

Southeast University



Submitted by

Labonna Saha Joyeta-2026000000271

Md . Sohan Sheikh-2026000000287

Emtahanul Islam Enan-2026000000296

INTRODUCTION

Cyber security refers to the practice of protecting systems, networks, and data from digital attacks. In today's digital world, where everything is connected through the internet, cyber security has become a crucial part of our daily life. From personal information to financial data, everything needs protection from cyber threats.

Main Discussion

Why Cyber Security Matters

Cyber Security is important for several reasons:

Protects personal information from theft.

Prevents financial fraud and cybercrime.

Secures business and organizational data.

Ensures safe online communication.

Protects national infrastructure and security.

Without proper cyber security measures, individuals and organizations may face serious financial and reputational damage

Types of Cyber Threats

1. Malware

Malware is malicious software designed to damage or gain unauthorized access to computer systems.

2. Phishing

Phishing attacks attempt to steal sensitive information through fake emails, messages, or websites.

3. Hacking

Hacking involves unauthorized access to computer systems or networks.

4. Ransomware

Ransomware locks a user's files and demands payment to restore access.

5. Data Breaches

A data breach occurs when confidential information is accessed or disclosed without authorization.

Protecting Personal Data

The following practices help protect personal data:

- Use strong and unique passwords.
- Enable two-factor authentication (2FA).
- Regularly update software and operating systems.

- Avoid suspicious links and attachments.
- Use antivirus and firewall protection.
- Backup important files regularly.

Who Are the Attackers?

- Cyber attackers may include:
 - **Individual Hackers**
 - Individuals seeking financial gain, recognition, or personal interests.
 - **Cybercriminal Groups**
 - Organized groups conducting large-scale cybercrime activities.
 - **Nation-State Actors**
 - Government-sponsored groups targeting other countries for political or strategic purposes.
 - **Insider Threats**
 - Employees or authorized users who intentionally or accidentally compromise security.

The Human Factor

Human error is one of the leading causes of cyber security incidents. Common mistakes include:

- Weak passwords
- Clicking suspicious links
- Sharing sensitive information
- Ignoring security updates

Therefore, user awareness and education are essential components of cyber security.

Digital Security is Everyone's Responsibility

Cyber security is not only the responsibility of IT professionals. Every internet user plays a role in maintaining digital security. By following safe online practices, individuals can significantly reduce cyber risks.

The Role of Antivirus and Firewall

Antivirus

Antivirus software detects, prevents, and removes malicious software from devices.

Firewall

A firewall monitors network traffic and blocks unauthorized access to systems.

Together, these tools provide an important layer of protection against cyber threats.

Future of Cyber Security

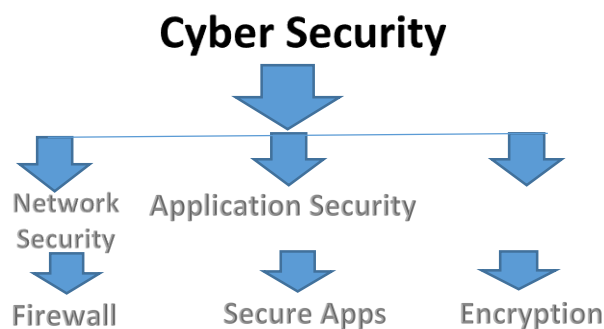
The future of cyber security includes:

- Artificial Intelligence (AI)-based threat detection.
- Advanced encryption technologies.
- Cloud security improvements.
- Increased cybersecurity awareness programs.
- Stronger regulations and compliance requirements.

As technology evolves, cyber security strategies must continue to adapt to emerging threats.

Diagram / Figure

Figure 1: Basic Cyber Security Framework

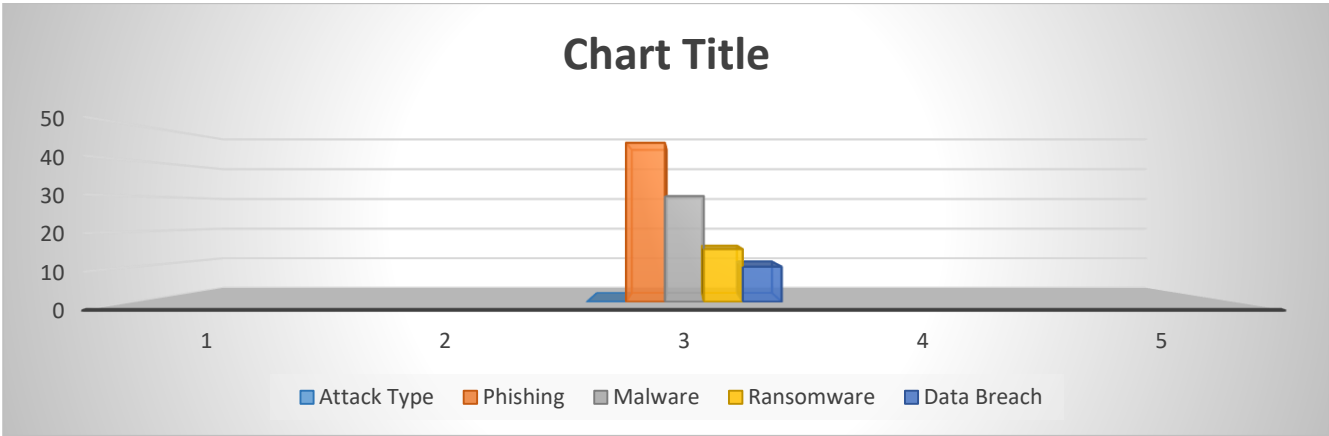


This diagram represents a Basic Cyber Security Framework .It consists of three main components.

Data Table

Attack Type	Number of Incidents
Phishing	45
Malware	30
Ransomware	15
Data Breach	10

Chart (Bar Chart)



Explanation of The Chart

The bar chart illustrates the number of incidents for different types of cyber attacks. Phishing attacks recorded the highest number of incidents (45), indicating that they are the most common cyber threat. Malware attacks ranked second with 30 incidents, while ransomware and data breach incidents were comparatively lower at 15 and 10 respectively. The chart highlights the need for stronger protection against phishing attacks through user awareness and security measures

Conclusion

Cyber Security is essential for protecting digital systems, networks, and sensitive information from cyber threats. By using strong passwords, security software, regular updates, and staying aware of online risks, individuals and organizations can create a safer digital environment. Cyber security is everyone's responsibility.

Contribution of Team Members

This project was completed through the collaborative efforts of all team members. Each member contributed in different areas to successfully complete the cyber security presentation.

Labonna Saha Joyeta (2026000000271) was responsible for preparing the report, creating the data table in Microsoft Excel, and designing the bar chart. She also assisted other team members whenever needed.

Md. Sohan Sheikh (2026000000287) conducted detailed research on the topic and prepared the presentation slides based on the collected information.

Emtahanul Islam Enan (2026000000296) contributed by creating some of the presentation slides and settings up our GitHub account.

Overall, the teamwork and cooperation among all members ensured the successful completion of the project.

Use of AI Tools

AI tools were used to gather information, improve grammar, organize the report structure, and assist in generating content ideas. All AI-generated content was carefully reviewed, verified, and edited by the team before being included in the final report. AI was used solely as a supporting tool and did not replace the team's original work.

References

- 1.Cybersecurity and Infrastructure Security Agency (CISA).
- 2.National Institute of Standards and Technology (NIST).
- 3.Cisco Cybersecurity Resources.
- 4.IBM Security Learning Center.